

Open WebUI

Open WebUI

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Users who are using the AI large model image or the solid-state factory image have already set up the environment. All setup-related steps in this tutorial can be skipped, and you can proceed directly to the "Running" section.

Demo Environment

Development Board: Raspberry Pi

SD (TF) Card: 64G (16G or above required; larger capacity allows you to experience more models)

Raspberry Pi 5 (16G RAM): Can run models with 14B parameters or less
Raspberry Pi 5 (8G RAM): Can run models with 8B parameters or less
Raspberry Pi 5 (4G RAM): Can run models with 3B parameters or less
Raspberry Pi 5 (2G RAM): Can run models with 1B parameters or less
Raspberry Pi 5 (1G RAM): Cannot run offline large models. Please refer to the online large model examples.

Open WebUI is an open-source project designed to provide a simple and user-friendly interface (UI) for managing and monitoring open-source software and services.

When using Open WebUI, there is a high probability of experiencing unresponsive conversations or timeouts. You can try restarting Open WebUI or switching to the Ollama tool to run the model!

Environment Requirements

For installing Open WebUI on the host machine and Conda: Node.js >= 20.10, Python = 3.11 is required:

Environment Setup Method	Difficulty (Relative)
Host Machine	High
Conda	Medium
Docker	Low

This tutorial demonstrates installing Open WebUI using Docker.

Docker Setup

Installing Docker via Official Method

If Docker is not installed, you can use a script for one-click Docker installation.

Docker is already installed in the Raspberry Pi 5 image we provide - no need to install it yourself!

Domestic users may be unable to install Docker via official methods. We recommend using Aliyun (Alibaba Cloud) to install Docker or simply use our image.

- Update the local package list

```
sudo apt update
```

- Upgrade installed packages

```
sudo apt upgrade
```

- Download and run the script

Download the get-docker.sh file and save it in the current directory.

```
sudo apt install curl
```

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

Run the get-docker.sh script file with sudo privileges.

```
sudo sh get-docker.sh
```

Installing Docker via Aliyun (Alibaba Cloud)

If you cannot install it yourself, please use the image we provide.

- Update the local package list

```
sudo apt update
```

- Install required software

```
sudo apt install apt-transport-https ca-certificates curl gnupg2 lsb-release software-properties-common
```

- Add GPG key for the software source

```
curl -fsSL https://mirrors.aliyun.com/docker-ce/linux/debian/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-archive-keyring.gpg
```

- Add Aliyun mirror software source

```
echo "deb [arch=arm64 signed-by=/usr/share/keyrings/docker-archive-keyring.gpg] https://mirrors.aliyun.com/docker-ce/linux/debian bookworm stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

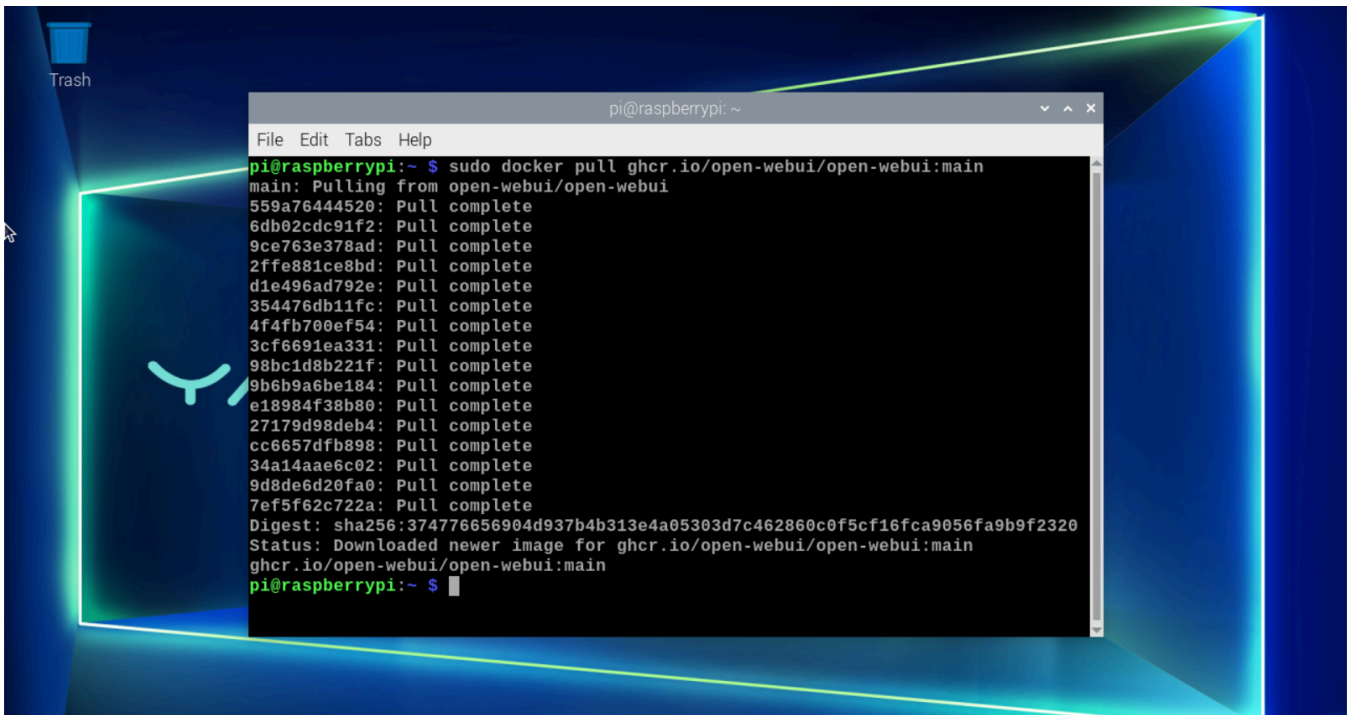
- Install Docker

```
sudo apt update
sudo apt install docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

Open WebUI Installation

For systems with Docker already installed, you can directly enter the following command in the terminal:

```
sudo docker pull ghcr.io/open-webui/open-webui:main
```



Running Open WebUI

Enter the following command in the terminal to start the specified Docker container:

```
sudo docker run --network=host -v open-webui:/app/backend/data -e
OLLAMA_BASE_URL=http://127.0.0.1:11434 --name open-webui --restart always ghcr.io/open-
webui/open-webui:main
```

```
pi@raspberrypi: ~  
File Edit Tabs Help  
pi@raspberrypi:~$ sudo docker run --network=host -v open-webui:/app/backend/data -e OLLAMA_BASE_URL=http://127.0.0.1:11434 --  
name open-webui --restart always ghcr.io/open-webui/open-webui:main  
Loading WEBUI_SECRET_KEY from file, not provided as an environment variable.  
Generating WEBUI_SECRET_KEY  
Loading WEBUI_SECRET_KEY from .webui_secret_key  
INFO: Started server process [1]  
INFO: Waiting for application startup.  
INFO: Application startup complete.  
INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)  
/app  
  
v0.3.5 - building the best open-source AI user interface.  
  
https://github.com/open-webui/open-webui  
  
INFO:apps.openai.main:get_all_models()  
INFO:apps.ollama.main:get_all_models()  
INFO: 127.0.0.1:55444 - "GET /health HTTP/1.1" 200 OK
```

After successful startup, access the following URL in your browser:

http://localhost:8080/

Administrator Account

On first use, you need to register an account yourself. This account will be the administrator account. You can fill in the information as required!

```
Since all content in our image has been set up and tested, you can directly log in using the account we registered:  
Username: yahboom  
Email: yahboom@163.com  
Password: yahboom
```

```
Username: yahboom
Email: yahboom@163.com
Password: yahboom
```

Registration and Login

Open WebUI - Chromium

Open WebUI

localhost:8080/auth/

oi

Sign in to Open WebUI

Email

Password

Sign in

Don't have an account? [Sign up](#)

Open WebUI - Chromium

Open WebUI

localhost:8080/auth/

oi

Sign up to Open WebUI

Open WebUI does not make any external connections, and your data stays securely on your locally hosted server.

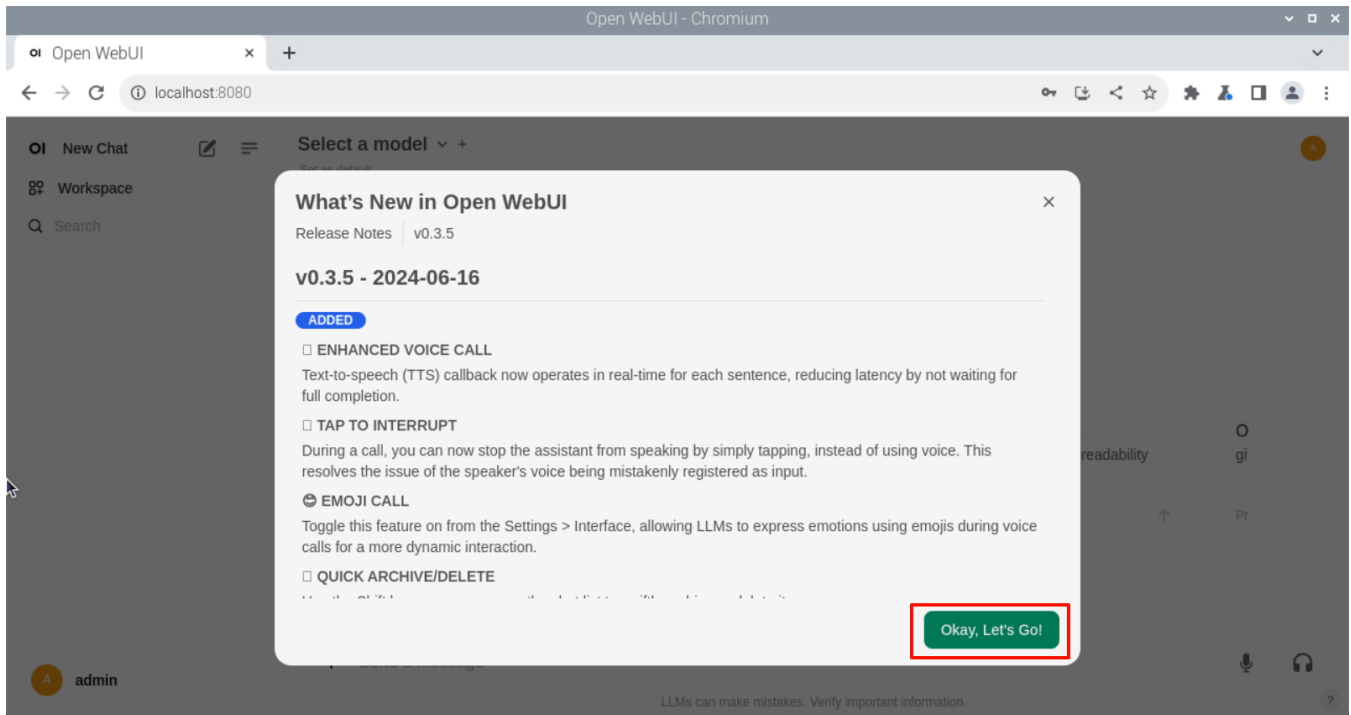
Name

Email

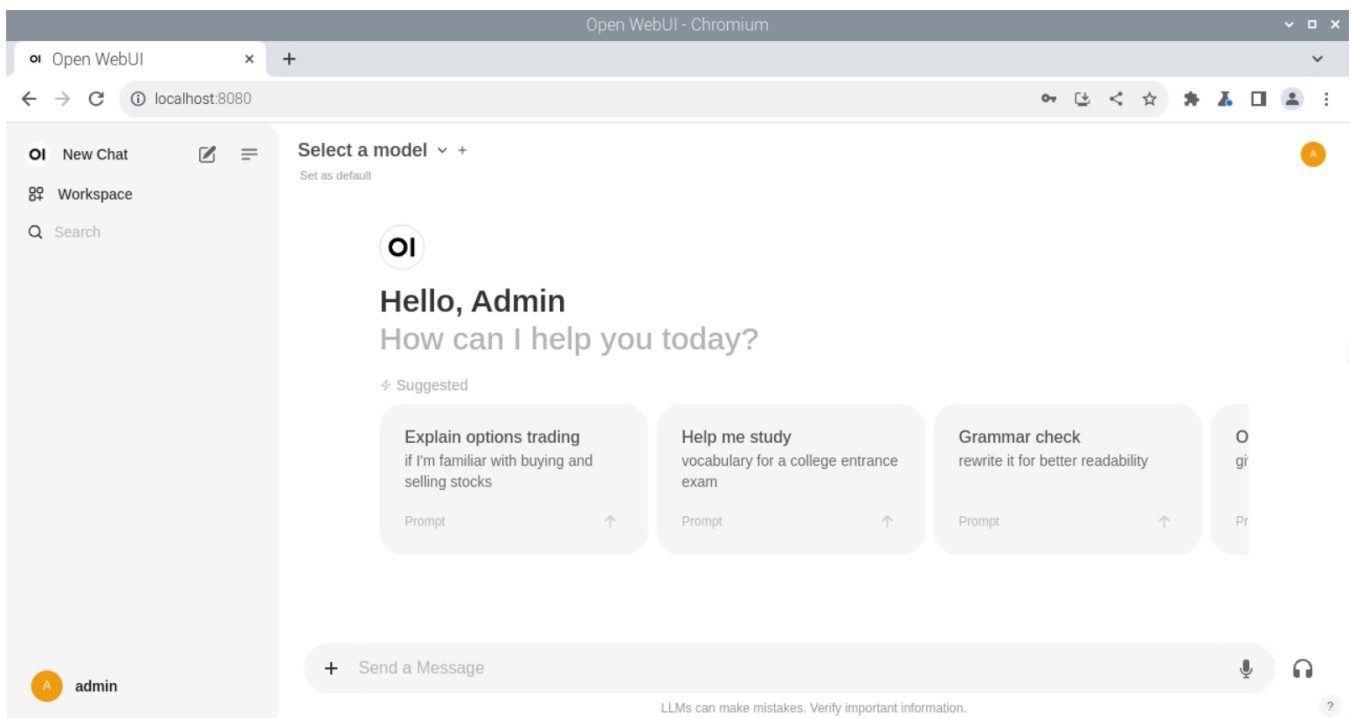
Password

Create Account

Already have an account? [Sign in](#)



Interface Overview



Model Chat

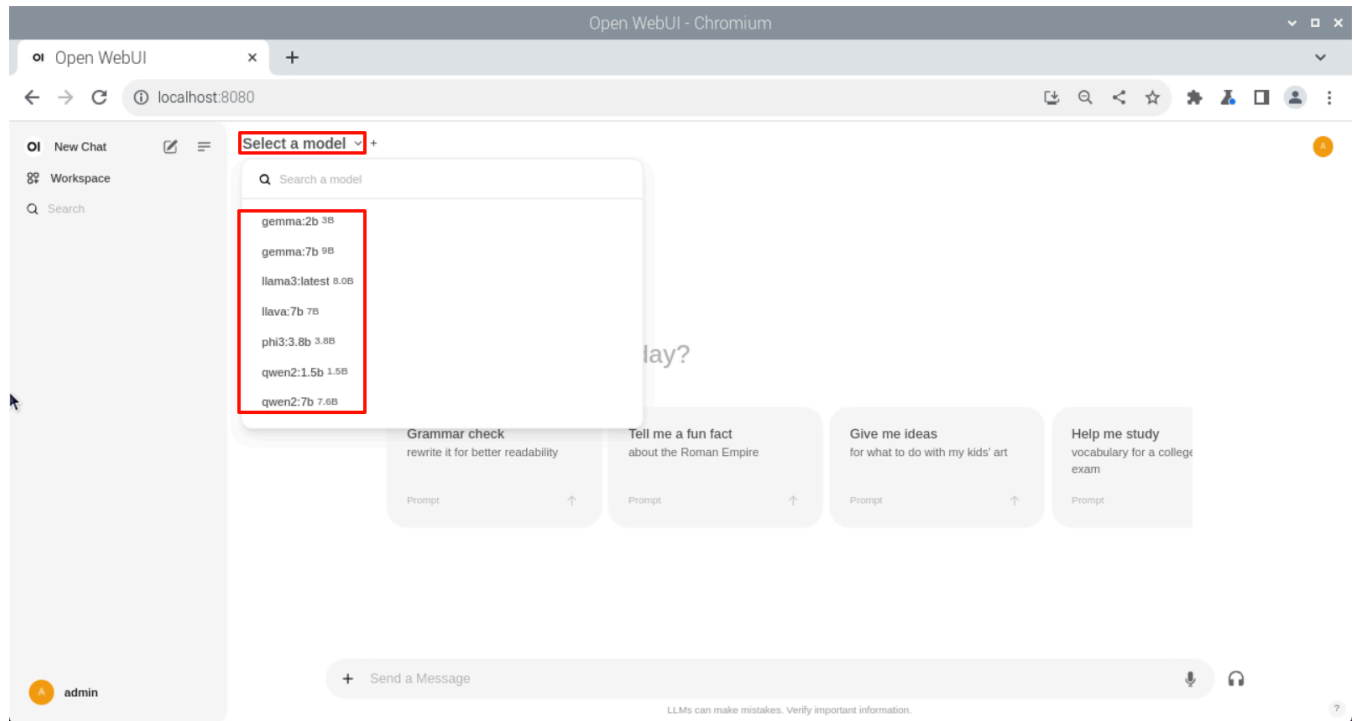
Using Open WebUI for chat will be slower than directly using the Ollama tool to run the model. It may even experience timeout and service connection failures. This is related to the Raspberry Pi 5's memory and cannot be avoided!

Users who are interested can switch to other Linux environments to set up ollama and open webui for chat

Switching Models

Click `Select a model` to choose a specific model for chat.

Models pulled using ollama will automatically be added to the Open WebUI model options. Just refresh the webpage and the new model will appear!



FAQ

Stopping Open WebUI

Stop Open WebUI from auto-starting.

- View running Docker containers

```
docker ps
```

- Stop a running Docker container

```
docker stop [CONTAINER ID] # Example: docker stop 5f42ee9cf784
```

- View stopped containers

```
docker ps -a
```

- Remove stopped containers

```
docker rm [CONTAINER ID] # Example: docker rm 5f42ee9cf784
```

Remove all stopped containers:

Common Errors

Cannot Start Open WebUI

- docker: Error response from daemon: Conflict. The container name "/open-webui" is already in use by container "cfc05c84f8e38b290337e7178c76fd1c49076f94b11ed3d49d9448be72b7f20f". You have to remove (or rename) that container to be able to reuse that name.

Solution: Stop Open WebUI once and restart it!

Service Connection Timeout

- Open WebUI: Server Connection Error

Stop Open WebUI once and restart it, then try asking again or switch to the Ollama tool to run the model and ask questions!